# Jordan Canonical Form: Answer Key

Problem sheet for the Day 11 class · CSIR NET / GATE previous year questions and exam-pattern problems

## Answers

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | (D) | (A) | (C) | (B) | 2 | (A),(C),(D) | (B),(C) | (B),(C) | (A),(B),(C) | (A),(C),(D) |

## Hints and short solutions

1. **(D)** GATE 2008 $T^3$ kills every basis vector ($e_1 \to e_2 \to e_3 \to 0$, $e_4 \to e_2 \to e_3 \to 0$), so $T$ is nilpotent. The image is $\mathbf{span}\{e_2, e_3\}$: rank 2, so not invertible; $T^2 e_1 = e_3 \neq T e_1$.
2. **(A)** GATE 2013 In Jordan form $T$ has blocks $J_k(0)$, $k \leq n+1$. $\mathbf{rank}\, T^n$ is the number $b$ of blocks of size $n+1 \geq 2$, so $2b \leq b(n+1) \leq m$ and $\mathbf{rank} \leq m - \mathbf{rank} = \mathbf{nullity}$. A nilpotent has trace 0, is not diagonalizable unless 0, and $n = m$ is impossible ($n+1 \leq m$).
3. **(C)** Exam pattern A nilpotent matrix is determined up to similarity by its Jordan block sizes, a partition of 5: $p(5) = 7$.
4. **(B)** Exam pattern Three blocks for 2 of total size 4 and largest size 2: $J_2(2) \oplus J_1(2) \oplus J_1(2)$. For 1, two $1 \times 1$ blocks. $A - 2I$ has rank 1 on $J_2(2)$, 0 on the $J_1(2)$'s and 2 on the eigenvalue-1 part: total 3.
5. **2** Exam pattern The blocks for 1 have largest size 2 and those for 2 all have size 1, with total size 4. Multiplicity of 1 equal to 2: $J_2(1) \oplus \mathbf{diag}(2,2)$. Multiplicity 3: $J_2(1) \oplus J_1(1) \oplus (2)$. So 2 classes.
6. **(A),(C),(D)** CSIR NET Dec 2015 $\chi_A = (x-2)^2(x-3)$ and $\mathbf{rank}(A - 2I) = 2$, so the eigenvalue 2 has one block: $J_A = J_2(2) \oplus (3) = B$. The eigenvalues are rational, so the similarity can be taken over $\mathbb{Q}$. The minimal polynomial is $(x-2)^2(x-3) = \chi_A$, and $A$ is not diagonalizable.
7. **(B),(C)** CSIR NET June 2016 $\chi_T = x^n$: 0 is the only eigenvalue, with algebraic multiplicity $n$. For $n \geq 2$, (A) fails, and (D) fails for $J_n(0)$ (the only diagonalizable nilpotent is 0). The official key accepts (C) or (B),(C).
8. **(B),(C)** CSIR NET June 2017 $T(1) = 0$, $T^4 = 0$: $T$ is nilpotent of index 4, so its Jordan form is the single block $J_4(0)$. It is not invertible and not diagonalizable in any basis.
9. **(A),(B),(C)** Exam pattern $\mathbf{rank}\, N = 4 - (\mathbf{number\ of\ blocks})$, so there are 2 blocks: sizes $3+1$ or $2+2$. Then $N^3 = 0$. For $2+2$, $N^2 = 0$ (nullity 4); for $3+1$, $\mathbf{rank}\, N^2 = 1$ (nullity 3). Nullity 2 never occurs.
10. **(A),(C),(D)** Exam pattern The minimal polynomial divides $x^2(x-1)$ but not $x(x-1)$, so it is $x^2$ or $x^2(x-1)$: there is a $J_2(0)$ block and $A$ is not diagonalizable. $J_2(0) \oplus (1)$ is not nilpotent. In every case $A^2$ is $0$ or $0 \oplus (1)$, and $A \neq 0$.